

Algorithm (ทำงานเร็ว / ประหยัดพื้นที่)

① ก. องค์ประกอบ Algorithm [เวลา (น้อย = ดี) / หน่วยความจำ (ใช้อย่างคุ้มค่าไหม?)]

② ประสิทธิภาพ $f(n)$

• Linear Loops $\oplus \ominus$

Ex) for (i=0; i<n; i++) $\rightarrow f(n) = n$

Ex) for (i=0; i<n; i+=2) $\rightarrow f(n) = n/2$

• Logarithmic Loops $\otimes \ominus$

Ex) for (i=1; i<n; i*=2) $\rightarrow f(n) = \log_2 n$

• Nested Loops

• Linear Logarithmic

```
for (i=0; i<10; i++) {
  for (j=1; j<10; j*=2) {
    application code
  }
}
```

Iterations: $f(n) = n \log_2 n$

Handwritten notes: $\log_2 10 = 4$ (approx), $10 \times 4 = 40$ iterations.

• Quadratic

```
for (i=0; i<10; i++) {
  for (j=0; j<10; j++) {
    application code
  }
}
```

Iterations: $f(n) = n^2, n \times n$

Handwritten notes: $10 \times 10 = 100$

NESTED LOOPS (CONT.)

• Dependent Quadratic

```
for (i=1; i<=10; i++) {
  for (j=0; j<=i; j++) {
    application code
  }
}
```

Iterations: $f(n) = \frac{n(n+1)}{2}$

i	j
1	1
2	1, 2
3	1, 2, 3
4	1, 2, 3, 4
5	1, 2, 3, 4, 5
6	1, 2, 3, 4, 5, 6
7	1, 2, 3, 4, 5, 6, 7
8	1, 2, 3, 4, 5, 6, 7, 8
9	1, 2, 3, 4, 5, 6, 7, 8, 9
10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Handwritten notes: $1+2+3+4+5+6+7+8+9+10 = 55$